

Gendered paths to technology leadership

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Results indicate that more than men, women aspiring technology leaders are heeding expert advice by assuming roles that focus less on technology and more on business and change management. The stereotype that women are better at working with people may be working in their favour in preparing them for advancement.

Introduction

The lack of women in the science, technology, engineering and mathematics occupations has been a concern for years (Cooper, 2000; Acker, 2004; Powell *et al.*, 2007; Boushey, 2009). While there has been a push to include more women in the male-dominated fields of information technology (IT) and high technology, the results have had limited success (Frenkel, 2008; Rosenbloom *et al.*, 2008; Boushey, 2009).

In this study, we focus on different career strategies employed by men and women who aspire to be leaders in the broadly defined field of technology, which for this study includes IT and related communication technologies. Examining the quest and path to leadership is particularly important in the field of technology since women have few role models and mentors to assist their career aspirations in a male-dominated environment (Acker, 1990; 2004). Women's leadership studies suggest that there is a greater chance of promoting women when there are women at the top of the organisation (e.g. Adams, 2010; Matsa and Miller, 2011). Indeed, more than one woman is needed since being a token woman can be associated with social status devaluation (McDonald *et al.*, 2004). Studies are needed to determine whether women are positioning themselves appropriately for advancement to technology leadership. Research in this area can clarify whether or not women are employing effective career strategies in this broad but important field.

Gender and technology leadership

As noted above, studies of women in technology fields show that women are under-represented. In the United States, for example, women make up half of the present labour force; only 20 per cent of engineering professions are female, 27 per cent are environmental scientists, 31 per cent are chemists, and 27 per cent are in computer and mathematical occupations (Rosenbloom *et al.*, 2008).

These statistics in technology careers and in leadership positions have been offered by stereotyping, discrimination, ability differences, socialisation techniques, child-

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rearing career interruptions and choice (Yap, 1995; Xie and Shauman, 2003; Aziz, 2004; Catalyst, 2004; Webster, 2004; King *et al.*, 2005; Mainiero and Sullivan, 2005; Castaño and Webster, 2011). Other studies attribute the low representation of women in technology to differences of interests, personality, work-family stress, compensation needs and organisational leadership (Loogma *et al.*, 2004; Broadbent and Kitzis, 2005; Catalyst, 2005; Litsky and Greenhaus, 2007; Castaño and Webster, 2011). Frenkel (2008), for example, found that the masculine culture in Israel's high-technology industry coupled with Israel's family devotion culture is a balancing act for the women encouraged to work full-time in high technology for the good of the nation. Use of work-family resources is accepted as a way to welcome women to the field but is still seen as a female resource that men tend not to use, even if they have wives who work full-time. A conclusion from these studies is that technology careers and leadership positions are relatively rare for women compared with men. Moreover, women have more complicated paths to leadership in the male-dominated environment (Eagly and Carli, 2007).

Other studies indicate that similarities of men's and women's professional needs and demands from organisations raise important issues for organisations with regard to cultures and internal environments, and to recruiting and retaining valued leaders (Powell *et al.*, 2006; Metz and Simon, 2008). For example, Metz and Simon (2008) argue that social trends such as the decrease in family size, the increase in the number of alternative family patterns such as single-parent households (Schneer and Reitman, 2002), the likelihood that women in senior management have fewer family responsibilities than men (Lyness and Thompson, 1997; Hewlett, 2002) and the increasingly more egalitarian approach to the division of labour in Western societies suggest that women and men in high levels of management should be more similar than different in their views of work and in their work experiences. Metz and Simon (2008) propose a focus on gender similarities (FGS) approach to studies which is in line with Catalyst's (2003) finding that women seek and can handle the challenges of executive leadership in similar numbers as men. While these societal trends promise women greater access to higher-level technology positions that require 'feminine' skills, Roan and Whitehouse (2007) found little optimism from the findings of their research, citing poor career advancement by women. A potential explanation is offered in the work of Webster (2004) who suggests that women are not accessing informal learning arrangements that are supplanting formal learning needed for advancement.

We surmise that women who are and aspire to be technology leaders may be affected by the range of the factors summarised above, including the quest for upward mobility, meaning in their work, and the opportunity to make a difference in changing technology professions. However, most of the previous studies mentioned above examined gender differences in choosing or staying in technology fields. We propose a merging of theoretical approaches to examine both gender similarities and differences in career strategies for advancement to technology leadership. This study focuses on those who have already chosen the field and desire to advance to higher levels of technology leadership. We seek to understand the nature of activities and preferences of men and women who are working in traditionally male environments and positions.

Changing demands of technology leadership

Changes in the work environment imply perceived changes in employees' careers (Hall, 2002: 15). Prominent researchers (e.g. Broadbent and Kitzis, 2005) and observers of professions (e.g. Melymuka, 2002) argue that technology leaders, chief information officers (CIOs) in particular, are at a crossroads with regard to the direction of and role in their careers. The position of the CIO, they suggest, is changing from a 'chief technology mechanic' role to competencies focused more on strategic business issues (Broadbent and Kitzis, 2005: 2).

Jeanne W. Ross, Director and Principal Research Scientist at the Center for Information Systems Research at the MIT Sloan School of Management, stated in her recent research (Ross, 2009) that firms are recognising the need for new skills. Two of these skills that are relevant to this article include business management and change management. Specifically, organisational teams that include technology professionals are expected to focus their efforts on business management issues such as cost containment and revenue production as well as change management initiatives to keep the firm competitive.

The changing nature of technology leadership roles

According to Hall, '... the *career role* is the expectations people hold for individuals in that career' (Hall, 2002: 72). Along those lines, if changing dynamics in the workplace are imposing greater expectations on technology leaders, then the nature of the career technology leader role changes. We take the traditional careers studies definition of 'career' as the series of positions and occupations one fills (*cf.* Hall, 2002; Baruch, 2004).

Pressures from globalisation, hypercompetition, economic instability and industry consolidation are changing requirements and roles of technology leaders. The evolving nature of the CIO position suggests the need for a change in career strategies to reach higher positions in this profession. This change is akin to changes in the way individual contributors in different functional areas of an organisation transition to organisational management, taking on multiple roles (Katz, 1974; Mintzberg, 1980). Technology managers are increasingly seen as organisational managers first and technology experts second.

Broadbent and Kitzis (2005) and others emphasise the need for business and strategic competencies in addition to technological expertise for success in the CIO role, in particular. Their conclusion is based on the notion that the CIO job is more 'executive' in nature than 'functional', as Stephens and colleagues (Stephens *et al.*, 1992) also found in their in-depth study of the work of five CIOs.

The position of the highest technology leader (CIO) has never been static since its inception. The evolution of CIO roles has been characterised in four waves: The first wave pictured CIOs as 'glorified data processing managers'. CIOs were technocrats in the second wave and business executives in the third wave. Now in the fourth wave, CIOs are being described as technocrats and business executives (Andrews and Carlson, 1997; Melymuka, 2002).

Technology leaders' competencies have evolved with their different roles. Gottschalk (2001) described nine CIO leadership roles: informational, decisional, interpersonal, chief architect, change leader, product developer, technology provocateur, coach and chief operating strategist, of which, the informational and change leader roles were the most important. More recent technology leaders have been responsible for strategic alignment between technology and business (Weiss and Anderson, 2004). Bassellier *et al.* (2001) described a set of core competencies that business managers needed in order to work more effectively with technology professionals. These included IT applications, system development, management of technology and access to technology (Bassellier *et al.*, 2001). Tacit knowledge components (interpersonal communication and cognition) are also required for business managers to work more effectively with technology professionals.

Finally, Broadbent and Kitzis (2005) describe 10 top priorities for today's CIOs. None of the 10 priorities deal with technical expertise but rather describe dimensions of supply-side leadership (mostly management behaviour competencies) and demand-side leadership (mostly business executive competencies).

Three roles in this study

Based on previous studies, we propose three broad technology leader roles that are important to higher-level technology leaders as noted by Pervan (1998). Pervan

found that chief executive officers (CEOs) consider the following issues as critical: technology management (managing and measuring the effectiveness of technology infrastructure and disaster recovery), strategy management (business process redesign, competitive advantage and information architecture) and people and support management (organisational learning and executive and decision support). Likewise, Lee (2002) found that assuming these roles can be part of career strategies for technology professionals.

Markus and Benjamin (1996) highlighted the importance of effective change management for information system (IS) specialists. They cite three reasons why IS specialists need to be effective change managers: (1) IS systems are organisational interventions; (2) the task is becoming more management than production with much of the work outsourced; and (3) organisational credibility is enhanced when dealing with the needs of the whole organisation.

Related to Pervan's and the previous studies cited earlier, the three roles we use in this study to determine how technology leaders are spending their time, expertise and preferred competencies are: (1) Technologist; (2) Change Agent; and (3) Business Expert. These have distinct and complementary competencies. Table 1 was created from interviews with four CIOs and the literature cited earlier. It illustrates the options for developing breadth and depth one can take in enacting a career as a technology leader. The Technologist focuses on the traditional functional aspect of technology work similar to what McKinsey termed an 'I-shaped career' with deepening expertise and impact in a specified field (Nanda and Morrell, 2002). Taking the McKinsey 'T-shaped' career approach, an individual broadens his or her scope of expertise by adding skills as a Business Expert and Change Agent while still retaining expertise in technology. In the T-shaped career, one would advance both deeper and broader but not as deep in technology as the Technologist who primarily develops expertise in the technology area.

Thus, the path to technology leadership is self-defined and influenced by a number of factors, including organisational expectations. The individual also self-defines what it means to him or her to be a technology leader. In this sense, 'leader' is a subjective term referring to one's identity rather than a specific position. The position of a leader is referred to as a subjective measure of a career (Baruch, 2004), whereby the individual

Table 1: Technology leader (CIO) career profiles: breadth and depth of competencies

(Depth)	Developmental areas of competency (breadth)		
	Technologist	Change agent	Business expert
Basic level	e.g. understanding existing systems and technology	e.g. understanding sources of conflict and resistance to change	e.g. understanding business practices, processes and approaches
	e.g. applying procedures, tools, and methods; designing and developing applications	e.g. focusing on results; building relationships and teamwork	e.g. managing projects; planning, prioritising and administering work
	e.g. designing technical architecture; integrating systems	e.g. leading, inspiring and building trust; principled negotiating	e.g. leading cross-functional technology and business teams
Expert level			

sets his or her own standards for achievement relative to who he or she wants to be. This measure is more predictive of career-related behaviours such as turnover, work satisfaction and work withdrawal (*cf.* Hall, 2002; Baruch, 2004) and allows us to capture the personal views of individuals that are driving their career-related behaviours. A high-achieving CIO, for example, may not see herself as a technology leader until she is internationally recognised, while others may see themselves as earning the technology leader identity when they reach the CIO level.

Ashforth *et al.* (2000) discuss the segmentation or integration of roles as instrumental in the transition between roles and role identification (i.e. technology leadership, in this case). Integration of roles makes the transition less difficult and enables identity formation. Therefore, based on the changing expectation cited above, we anticipate that aspiring technology leaders (those who aspire the technology leader identity) desire an integrated role view of the technology leader (the T-shaped career), so that they meet the expectations that people hold for technology leaders. The new roles added to the Technologist require the social skills that women prefer more than men (Eagly and Carli, 2007), yet studies show that women who advance in leadership tend to adopt masculine styles (Eagly and Johnson, 1990; Vinnicombe and Singh, 2002; Powell *et al.*, 2009).

Little research exists that addresses whether men or women in the technology field are acknowledging these changing role trends differently and, if so, in what ways. The extant literature is not clear, so we leave the prediction of gender-related migration to the integrated roles as an open question to examine. To summarise, this exploratory study investigates whether men and women aspiring to be technology leaders (as self defined) are adjusting their roles and practices to align with industry experts' prognostications of professional changes needed to succeed in the field, thereby placing themselves in sync with the expectations of technology leaders.

Research questions and hypotheses

While exploratory in nature, this empirical study addresses three research questions focusing on the nature of technology leadership roles. First, 'How much time do aspiring technology leaders, men and women, currently spend in technical compared to business and change management roles?' Second, as individuals who desire to advance their careers as technology leaders, 'How much time do technology leaders, men and women, prefer to spend in these different roles?' Third, acknowledging the growing, albeit small, presence of women in the technology workforce, the central question is, 'Are aspiring women technology leaders similar to men in time spent, type of expertise used and expressed preference for particular roles?'

The first research question indicates whether or not and the extent to which the arguments discussed earlier were observed with our respondents: that is, 'Is the nature of the roles and careers of technology leaders more strategic and less technologically oriented as they reach for technology leadership?' One would expect this to be the case, given the changes in the business environment, technology investments made to enhance the organisation's competitiveness and arguments stated earlier. If these arguments are valid, we would expect to find that aspiring-to-be technology leaders, men and women, would migrate their roles from the current state to those expected of technology leaders.

H1: Time spent in preferred roles over current roles should indicate:

- Less time spent as a Technologist;
- More time spent as a Business Expert;
- More time spent as a Change Agent.

Furthermore, in addressing the first research question, we would expect to see aspiring leaders, men and women, rate their levels of expertise in the leadership roles higher than in the Technologist role as they prepare for advancement (however they define leadership).

H2: Aspiring technology leaders will see themselves as having more expertise as a Business Expert and Change Agent than as Technologist.

To test the second and third research questions, we asked, 'Were there any significant gender differences in preferences for any particular one of these three roles?' Again, if the overall argument posed at the outset is valid, that technology leaders are becoming more strategic and less technical, we would expect the same or similar results for men and women because both genders would be trying to meet societal expectations for the technology leader role.

H3: Preferred time spent in roles should be the same for men and women.

Methodology

Background and procedure

Individuals were selected from a list of IT majors obtained from the alumni office of a business university in the northeastern United States. A web-based survey was emailed to 258 individuals who were identified as aspiring technology leaders based on their university major and response to the request that asked to survey aspiring technology leaders. The survey was designed after a literature review was completed and face-to-face interviews were conducted with current CIOs whose feedback helped create and refine the survey questions. The questionnaire included open-ended questions that invited respondents to provide their views of career advancement in the technology field. The open-ended questions provided a multi-method approach to examine the research questions in a qualitative manner to supplement and triangulate quantitative findings (Denzin and Lincoln, 1998; Tharenou *et al.*, 2007). There were 55 usable responses for a response rate of 21.3 per cent. The survey questions are provided in the Appendix. Note that data for testing hypotheses were derived from single-item questions so reliability and validity of measures are not presented here. However, we did verify understanding of the three role definitions by asking for examples of tasks that are associated with those roles.

Sample characteristics

Eighteen (32.7 per cent) of the 55 respondents were female and 37 were male, a realistic sample as 13.7 per cent of the highest-level CIOs in the Fortune 1,000 companies are women. The sample originates from organisations ranging in size from 40 to 70,000 employees [mean = 9,504; standard deviation (SD) = 17,837] with technology staffs ranging from 2 to 50,000 employees (mean = 2,110; SD = 9,096). Eighteen have undergraduate degrees, 16 have a Master of Business Administration degree and 21 have a master's degree in IT or computer information systems. Current job titles of the respondents who indicated a title were classified as business manager (e.g. product manager for an IT product; $n = 3$), technical specialist (e.g. programmer; $n = 16$), IT manager (e.g. director-level position; $n = 21$) and IT executive (e.g. vice president (VP) or CIO; $n = 14$), according to the individual's level in the organisation and job function.

In addition to the variables described earlier (gender, organisation size, staff size, education and job title), subjects provided an approximation of the per cent of time currently spent on the three areas of competency, an approximation of the per cent of time preferred to spend on the three areas of competency and self-assessments of expertise in each of the three competency areas measured by one item for each area of competency on a 9-point scale. Seven open-ended questions captured career path information, examples of job tasks related to the three competency areas and advice for aspiring technology leaders entering the workplace.

Findings

Descriptive findings

A correlation table with significance levels indicated is found in Table 2. Organisation size and technology staff size are positively related as would be expected (Table 2). Organisation size and staff size are positively related to the per cent of time that technology leaders reported they spent in their roles as Technologist—a finding that is consistent with the specialisation of tasks found in larger organisations. Technology expertise is positively related to reported per cent of time spent as Technologist ($r = 0.428, p \leq 0.001$) and negatively related to time spent as Business Expert ($r = -0.451, p \leq 0.05$). Technology expertise is also positively related to the amount of time preferred to be spent as Technologist ($r = 0.428, p \leq 0.01$). Other correlations indicate distinctions among the role dimensions. There were no significant correlations between gender and job title or educational level.

Hypothesis testing and quantitative analyses

Table 3 depicts men's and women's actual time spent in the three roles. Both men and women currently spend more time as a Business Expert than they do as a Technologist. *t*-tests reveal that women currently spend significantly more time as Business Experts (49 per cent) compared with men (36 per cent; $t = -2.286, p \leq 0.05$; Note: Men were coded as 1 and women as 2). Men spend about 13 per cent more of their time as Technologists than do women ($t = 2.261, p \leq 0.05$), but the same amount of time as Change Agents.

Table 3 also shows men's and women's preferred roles as they aspire for leadership. Both men and women prefer to spend the most time in a Change Agent role, followed by time as a Business Expert and the least amount of time in the Technologist role.

Men are higher in their self-reported technology expertise ($t = 3.304, p \leq 0.01$) and change management expertise ($t = 1.993, p \leq 0.05$) than women. There was no statistical difference in self-reported expertise as a Business Expert. Variances were determined unequal between men and women for the technology and business expertise according to Levene's test for equality of variances so adjusted *T*-statistics were used to determine differences in expertise by gender for these variables.

Hypothesis 1 focuses on the difference between the amounts of time currently spent in the three roles compared with preferred time in each role. Paired *t*-tests were used to examine this question (Sekaran, 2000). The only significant difference indicated was for a reduction in time spent in the Technologist role ($t = -2.534, p \leq 0.01$). The data show desired increases in time spent in the Business Expert and Change Agent roles, but the differences were not significant for a 95 per cent confidence interval. However, the desire for more time spent in the Change Agent role was marginally significant ($t = 1.892, p = 0.06$). These findings partially support Hypothesis 1.

For Hypothesis 2, paired *t*-tests were used to determine whether respondents felt they had more change management and business expertise than technology expertise. For both tests, technology expertise was significantly lower (business: $t = -2.746, p \leq 0.01$; change management: $t = -1.974, p \leq 0.05$), supporting Hypothesis 2.

While the percentages shown in Table 3 are interesting, there are no significant differences in aspirations by gender, supporting Hypothesis 3. Having noted this fact, both groups would prefer Business Expert roles: women's 50 per cent to men's 40 per cent. Change Agent roles are preferred next by men's 30 per cent to women's 29 per cent. The Technologist role is preferred least: by men's only 25 per cent to women's 21 per cent.

Looking at the means, SD and results of *t*-tests of the full sample and by gender in Table 3, note that the SD for the male sample are more uniform, and that there is a wider range in the women's sample.

Table 2: Correlations

	Organisation size	Staff size	Time as Technologist	Time as Business Expert	Time as Change Agent	Preferred time as Technologist	Preferred time as Business Expert	Preferred time as Change Agent	Technologist expertise	Business expertise	Change Agent expertise
Organisation size	1										
Staff size	0.555 ^a	1									
Time as Technologist	0.175	0.485 ^a	1								
Time as Business Expert	-0.149	-0.269 ^b	-0.602 ^a	1							
Time as Change Agent	-0.006	-0.233	-0.330 ^b	-0.305 ^b	1						
Preferred time as Technologist	0.001	0.186	0.727 ^a	-0.489 ^a	-0.172	1					
Preferred time as Business Expert	-0.275 ^b	-0.164	-0.311 ^b	0.648 ^a	-0.270	-0.488 ^a	1				
Preferred time as Change Agent	0.261	-0.003	-0.367 ^a	-0.201	0.450 ^a	-0.408 ^a	-0.585 ^a	1			
Technologist expertise	-0.069	0.142	0.612 ^a	-0.451 ^b	-0.129	0.428 ^a	-0.254	-0.159	1		
Business expertise	-0.143	-0.175	-0.019	0.100	-0.049	-0.202	0.052	0.140	0.282 ^b	1	
Change Agent expertise	0.157	0.050	0.080	-0.272 ^b	0.233	-0.129	-0.312 ^b	0.448 ^a	0.227	0.606 ^a	1

Notes: ^aCorrelation is significant at the 0.01 level (2-tailed).

^bCorrelation is significant at the 0.05 level (2-tailed).

Table 3: *Gender difference statistics*

Gender	Actual % time spent as Technologist	Actual % time spent as Business Expert	Actual % time spent as Change Agent	Preferred % time spent as Technologist	Preferred % time spent as Business Expert	Preferred % time spent as Change Agent	Technologist expertise	Business expertise	Change Agent expertise
<i>Male</i>									
Mean	33.51	36.22	29.03	25.14	39.72	34.86	6.68	6.95	6.97
Sample size	37	37	37	36	36	36	37	37	37
Std. deviation	18.366	17.417	14.929	13.861	17.401	18.497	1.6	1.268	1.404
<i>Female</i>									
Mean	20.56	49.72	29.72	21.88	50.00	29.41	4.67	6.67	6.00
Sample size	18	18	18	16	17	17	18	18	18
Std. deviation	22.938	25.979	19.514	26.638	26.161	22.212	2.326	2.275	2.196
<i>Total</i>									
Mean	29.27	40.64	27.91	24.13	43.02	33.11	6.02	6.85	6.65
Sample size	55	55	55	52	53	53	55	55	55
Std. deviation	20.692	21.345	16.434	18.517	20.924	19.715	2.077	1.649	1.745
T-statistic	2.261*	-2.286*	-0.567	0.583	-1.699	0.938	3.304**	0.485	1.993*

*Denotes $p \leq 0.05$, **Denotes $p \leq 0.01$; ***Denotes $p \leq 0.001$.

Qualitative findings

Seven open-ended survey questions provided qualitative data. These data were coded for consistency with the evolving view of an increased level of multi-role technology leadership for career advancement, and also analysed for thematic content. Our analysis found no comments refuting the need for greater attention to Business and Change Agent role behaviours. A review of the qualitative data collected suggests two themes (Miles and Huberman, 1994): (1) a transition of the field from focusing on the Technologist role to one of multiple roles; and (2) climbing the career ladder to senior management in technology involves business knowledge and people management.

Examples of the first theme (transitioning to multiple roles) presented below are consistent with quantitative findings supporting Hypothesis 1. These comments from both men and women show that the respondents feel that technology expertise alone is not the pathway to successful technology leadership.

Being a CIO/IT leader isn't about implementing the coolest technology—it is about implementing the most relevant technology to solve a business problem. (Female, Senior IT Manager, Retail Company)

Be ready and willing to make the move into a supervisory or technical leadership position, even if you are managing 1 or 2 staff. You won't attain a CIO or leadership role on tech skills alone. (Male, Associate VP of IT, Hospitality Company)

Major career advancement problems include

Possible perception that IT is primarily 'plumbing' within the organization. (Male, Senior Director of IT, Biotech Company)

You need to think more like a business person than a techie! (Male, CIO, Medical Device Company)

Examples of the second theme (business and people management attention) presented below are consistent with quantitative findings in support of Hypothesis 2. The emphasis on business and working with others for the betterment of the enterprise is evident in comments from both men and women.

Technology leaders are paid for knowing 'why', not for knowing 'how'. This is a big challenge for folks who come up the technical ladder. (Female, CIO, University)

Learn how to run a business. IT is just a business that provides technology services. (Male, Senior Director of Global Technology, Technology Company)

Alignment with the business is critical—a partnership must exist. IT must be viewed as a value-add and high ROI organization." (Male, IT Director, Insurance Company)

Career advice:

Play well on a team and learn how to connect to resources, rather than 'owning' them. (Female, CEO/President, Online Marketing Company)

Career problems include

Inability to lead select/identify staff who can move organization forward and to admit mistakes and change course rapidly. (Female, IS Manager, Insurance Company)

Both men and women offered comments demonstrating support of Hypothesis 3. The comments made by both men and women acknowledge the desire for integrated roles in conducting their work.

Conclusions, limitations and implications

The following conclusions from this preliminary study provide an important basis for further study and dialogue in theory development, recruiting, hiring, training and mobility assignments among aspiring technology leaders:

- Men and women aspiring to be technology leaders are more similar than different with regard to time spent, expertise used and preference for work roles.
- Men and women see the key to achieving leadership roles as the ability to link technology to business goals and people management activities.
- Women technology leaders are more apt to be in business roles than are men.
- Men prefer business roles, although they tend to be working as Technologists.
- Women seem satisfied with their roles, which are predominantly business focused.

In total, these exploratory findings are consistent with the recognition of changing expectations for the technology leader role as a more strategic business partner than a technology enabler. Furthermore, both men and women are embracing the change as indicated by their desire to meet the changing expectations. The importance of business and change management roles and competences is clearly endorsed by men and women in this study looking towards the leadership level of technology.

Aspiring technology leaders in our study expressed the desire to meet technology leader role expectations by assuming the two business leader roles more than as Technologists, as indicated in their qualitative comments and ratings of desired time in the three roles of technology leadership work. Yet, some in the study did not express the desire to completely delegate or forego the Technologist role. The large SD of women's ratings of preferred time spent in the technologist and business roles shows some apprehension of women to migrate towards the expected profiles. All told, however, the findings support the integrated approach to transitioning roles (Ashforth *et al.*, 2000).

In terms of contribution to the literature regarding gender-related role changes, this study's findings also support, in part, Metz and Simon's (2008) FGS approach of studying gender in management research that emphasises gender similarities rather than gender differences. In particular, this study is consistent with Catalyst (2003) and Metz and Simon's (2008) findings that men and women hold similar beliefs of achieving their aspirations in their current organisations and how to do so. This study also extends these previous studies in two substantial ways. First, it was conducted across organisations, while the Metz and Simon (2008) study was conducted in one organisation. Second, this study measured behaviours in addition to attitudes by capturing time spent in roles as well as desired time to be spent in roles. These findings suggest broadened applicability of the FGS approach.

This study also adds to the body of gender difference studies that investigate aspects of the traditionally male technology position characterised by the technical and, more recently, leadership nature of the tasks (*cf.* Hofstede, 1980; Tannen, 1990; Gray, 1992; Connell, 1995). While we find that women in this field are more similar than different than men in their preferences and views about what is expected of the technology leader role, it is particularly interesting that women tend to spend more time than men in business roles. Previous research cited earlier suggests that it is conceivable that men are being stereotyped as more appropriate for the Technologist role or women may be seen as having the people skills necessary for the leadership roles. We found few studies of favourable stereotypes for women battling male hegemony and thus see this topic as an under-researched area that should be pursued to offer more options for women such as what we found here.

Implications of this study for organisations suggest that men and women are adopting the T-shaped career trajectory, seeking more business involvement and Change Agent responsibilities along with their technology roles and expertise. These findings are consistent with the notion to integrate the roles when making role transitions and forming an identity (Ashforth *et al.*, 2000; Hall, 2002). For men seeking technology leadership, this advice is imperative since our findings suggest that women are filling the positions that provide business experience more so than men.

From this study, we conclude that the desire to meet the new role expectations and the desire to transition more easily to leadership is motivating these aspiring technology leaders to retain the technology component as they add more of the Business Expert and Change Agent roles. Satisfying these aspirations could enhance motivation

and performance as they see themselves moving closer to preferred role integration configurations. Aspiring technology leaders are likely to inquire about the integration of business and strategy with technology in the roles they assume before joining a company.

We acknowledge and caution that this study is based on correlations, not causation. The limited sample includes a range of technology leaders, from CIOs to team leaders, all of whom feel they are aspiring to become technology leaders. Continuing research is needed that focuses on differences that may be associated with leadership levels (CIOs, directors, chief technology officers, technology VPs) and with larger samples of men and women. Even so, we did control for organisational title in our analyses, so we expect our findings to be robust. Also, studies by industry and in other geographic areas would be informative. It is possible that the New England area where these data were collected has a more adaptable population given the high concentration of universities and early stage and high-technology companies in the region. Case studies that include more in-depth observations and more interviews could provide insight and clarification of the themes discussed here.

In summary, this study sought to understand whether current activities and preferences of technology leaders are consistent with the integrated roles that technology researchers suggest are necessary for success in their changing technology profession. We did observe this phenomenon occurring. This study also examined gender differences and found an interesting result, namely women may actually have a stereotype working in their favour. They are more likely than men to already perform Business Expert and Change Agent roles in addition to the Technologist role. Our results indicated this observation even when we accounted for education and organisational level. This finding suggests that leadership roles in traditionally male professions may also provide an opportune route for women aspiring leadership positions in technology fields if they can overcome other barriers associated with working in male-dominated fields and environments.

Based on the reported findings and literature, this study also supports a new approach for examining role changes from a gender-based perspective. We found differences in behaviours, possibly constrained by stereotyping that favours female collaboration skills, and also similarities in leadership aspiration attitudes. Unlike the findings of Webster (2004) and Powell *et al.* (2009), this study finds that women are positioning themselves for advancement by learning on the job to develop the skills for advancement. A reason that may explain why our findings differ from previous studies is that our sample is more current and from an area of the United States known for careers in technology and leadership training. Perhaps, time and local culture is having a positive effect in socialising women in the technology field and leadership. A logical extension of this study could combine gender similarity and difference approaches to examine both attitudes and behaviours of men and women. As more women in the workforce become the breadwinners of families (Boushey, 2009), women need to be aware of opportunities where they can find entrées into traditional male bastions. Moreover, these findings offer organisations important information for recruiting, mentoring, career management and retaining needed talent in the leadership pipeline of women and men, especially in technology.

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Appendix 1

Survey questions (conducted online)

Organisational setting

1. Organisation size (approximate number of employees) _____
2. Technology staff size (approximate number of employees) _____

Your role(s)

1. Think about the demands of your job in terms of the three areas describes below. What per cent of your time, energy and expertise do you spend in each of the following roles. (0–100%)
 - a. Technologist—focuses on development, design and maintenance of technology applications and systems _____%
 - b. Business Specialist—focuses on developing, analysing and solving business related problems and opportunities _____%
 - c. Change Agent—focuses on designing, implementing and leading technology/business transitions and transformations _____%
 - d. Total = _____100_____
2. Provide an example of a representative activity you perform in each of these three roles.
 - a. Technologist _____
 - b. Business Specialist _____
 - c. Change Agent _____
3. Thinking about the three roles (Technologist, Business Specialist, Change Agent) and your career aspirations and goals, **where would you like to spend your time, energy and expertise** (assuming additional experience and education may be required).
 - a. Technologist—focuses on development, design and maintenance of technology applications and systems _____%
 - b. Business Specialist—focuses on developing, analysing and solving business related problems and opportunities _____%
 - c. Change Agent—focuses on designing, implementing and leading technology/business transitions and transformations _____%
 - d. Total = _____100_____
4. How would an objective external evaluation panel rank your skills and expertise for each of the following areas? (1–3 = basic skills; 4–6 = above average; 7–9 = expert)
 - a. Technologist—focuses on development, design and maintenance of technology applications and systems _____
 - b. Business Specialist—focuses on developing, analysing and solving business related problems and opportunities _____
 - c. Change Agent—focuses on designing, implementing and leading technology/business transitions and transformations _____

Today's technology leader and trends

1. Identify 3–5 issues and inhibitors that currently exist (or are on the horizon) that could derail a professional aspiring to be a Technology Leader from succeeding?
2. Reflecting on your career path to Technology Leader, what would you say were the most helpful development/learning turning-points, experiences you had?
3. What developmental experiences and education do you now want and need to reach your career aspirations? _____
4. In what ways has your organisation hindered your career development and path?
5. In what ways has your organisation helped and facilitated your career development? _____
6. What are some of the ways that your organisation could help facilitate your career development? _____
7. What three pieces of wisdom/advice from your experience and knowledge do you have for people aspiring today to reach the Technology Leader level? _____
8. Please offer the email addresses or names and companies of two other technology executives, preferably CIOs, that you know who can help by completing this survey. _____

Personal background information

Age _____
Gender (check box—male or female) _____
Education (highest degree) _____
Current job title _____
Career path (list previous job titles) _____
Technology expertise (areas of proficiency and years of experience) _____
Business expertise (areas of proficiency and years of experience) _____